Asim Avhad

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Summary

Master of AI student at Monash University working across NLP, generative AI, deep learning and multi-agent planning. Recent solo work: a retrieval-augmented support assistant on a fine-tuned DistilBERT, a train scheduler with a 92% success rate on 152 trains (top 20 in cohort) and a top 15 Kaggle finish. Guest lecturer on CNNs, Transformers and GenAI. Based in Melbourne; available full-time from 18 January 2027.

Education

Master of Artificial Intelligence | Monash University, Melbourne, VIC Feb 2026 – Dec 2027 (expected)
- Semester 1 GPA 3.5/4.0 (92% in Python programming). Units: Deep Learning, Planning and Automated Reasoning

B.E. Computer Science and Design | K. K. Wagh Institute of Engineering Education and Research, India 2021 – 2025
- CGPA 8.35/10. Coursework: algorithms, machine learning, operating systems, computer networks

Projects

Support Ticket Copilot (NLP and Generative AI) | *Python, PyTorch, Hugging Face, RAG, Gemini, Streamlit* 2026
- Built a support assistant that classifies messages into 77 banking intents, retrieves help-centre sections with hybrid BM25 and embedding search (Reciprocal Rank Fusion), and has Gemini or a local Ollama model draft cited replies.
- Fine-tuned DistilBERT on Banking77 (13,083 queries) in a custom PyTorch loop with mixed precision, against a TF-IDF baseline at 91.0% accuracy; intent routing raised top-3 retrieval hit rate from 0.64 to 0.77.
- Added guardrails that hand uncertain or off-topic tickets to a human, escalate fraud and strip invalid citations; with the baseline classifier, a 0.4 confidence threshold automated 92% of tickets at 94.7% intent accuracy.

Galaxy Morphology Classification and Redshift Estimation | *Python, PyTorch, CUDA, Flask* 1st Place, 2025
- Trained CNNs on a GPU with CUDA to classify galaxies by shape and estimate their redshift from telescope images.
- Improved redshift estimates by feeding CNN features into an MLP, validated on a held-out set; served it through Flask.

Flatland Multi-Agent Train Scheduling | *Python, space-time A*, Large Neighbourhood Search* Solo, Monash, 2026
- Routed 152 trains with deadlines over a shared rail network using space-time A* with a reservation table, reverse-BFS distance heuristics and prioritised planning over 16 agent orderings.
- Improved plans with adaptive Large Neighbourhood Search over groups of 12 trains and replanned online after breakdowns; reached a 92% success rate, top 20 in the cohort.

Deep Learning Kaggle Competition | *Python, PyTorch, transfer learning* Solo, Monash, 2026
- Fine-tuned pretrained models, comparing architectures and settings on a validation split; top 15 in the cohort.

Experience

Guest Lecturer, Deep Learning | K. K. Wagh Institute of Engineering Education and Research Jun 2025 – Jan 2026
- Invited back several times to teach final-year B.E. students CNNs, image classification, Transformers and GenAI.

IT and Cybersecurity Intern | FIVE Holdings (hospitality group), Dubai, UAE Dec 2023 – Jan 2024
- Helped deploy ManageEngine UEM and MDM across a live hotel, enrolling devices into one secured inventory for IT.
- Applied endpoint security procedures on live systems and mapped network, POS and back-office data flows.

Teaching Assistant, MongoDB | K. K. Wagh Institute of Engineering Education and Research Nov 2023
- Ran hands-on MongoDB lab sessions for AI and Data Science students and helped them debug their own queries.

Technical Skills

- **Languages:** Python, C++, C, SQL, JavaScript, TypeScript
- **NLP and Generative AI:** Hugging Face Transformers, RAG, embeddings, prompt engineering, Gemini API, Ollama
- **Machine Learning:** PyTorch, TensorFlow, Keras, scikit-learn, CNNs, computer vision, OpenCV, CUDA, Pandas, NumPy
- **AI Planning and Search:** A* and heuristic search, multi-agent path finding, Large Neighbourhood Search, PDDL
- **Software and Tools:** Flask, Streamlit, Node.js, React, Next.js, REST APIs, MongoDB, MySQL, Docker, Git, pytest

Achievements and Additional

- **Competitions:** 1st Place, IEEE Technovation (Apr 2025) and MET Enthusia (Mar 2025) project competitions
- **Certifications:** Supervised ML (Stanford, DeepLearning.AI); Full Stack Web Development (Apna College)
- **Languages:** English (fluent), Marathi (native) | **Interests:** zonal-level swimmer, 10 Himalayan treks, flute